edX UTAustinX UT.ALA
Advanced Linear Algebra: Foundations to Frontiers

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Course | Progress | Dates | Discussion | Syllabus | Notes | Summer22 Cohort

Your progress

Course completion

This represents how much of the course content you have completed. Note that some content may not yet be released.

Grades

This represents your weighted grade against the grade needed to pass this course.

Your current grade

Passing grade

You're currently passing this course

Grade summary

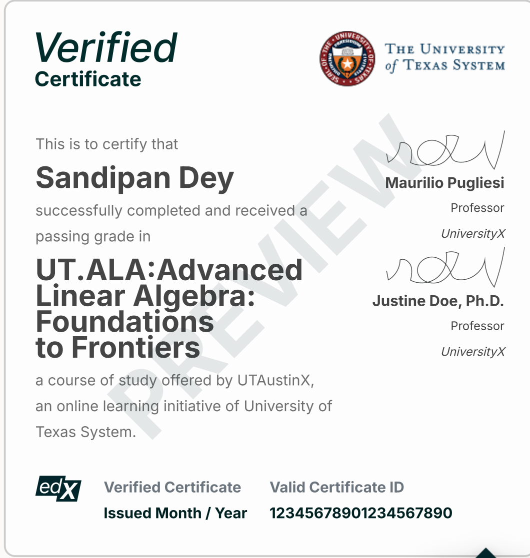
Assignment type	Weight	Grade	Weighted grade
Week 01	10%	100%	**10%**
Week 02 [1]	10%	100%	**10%**
Week 03	10%	100%	**10%**
Week 04	10%	100%	**10%**
Week 05	10%	100%	**10%**
Week 06	10%	100%	**10%**
Week 7-8	10%	100%	**10%**
Week 09	10%	100%	**10%**
Week 10	10%	100%	**10%**
Week 11	10%	100%	**10%**
Your current weighted grade summary			**100%**

[1]The lowest 2 Week 02 scores are dropped.

Detailed grades

- **Practice Scores:** Scores from non-graded activities meant for practice and self-assessment.
- **Graded Scores:** Scores from activities that contribute to your final grade.

Week 1: Norms	Score
1.1 Opening Remarks	6/6
1.2 Vector Norms	23/23
1.3 Matrix Norms	29/29



85%

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Related links

- Dates
  A schedule view of your course due dates and upcoming assignments.
- Course Outline
  A birds-eye view of your course content.

100%

Week 1: Norms	Score
1.4 Condition Number of a Matrix	7/7

Week 2: The Singular Value Decomposition	Score
2.1 Opening Remarks	3/3
2.2 Orthogonal Vectors and Matrices	33/33
2.3 The Singular Value Decomposition	19/19

Week 3: The QR Factorization	Score
3.1 Opening Remarks	3/3
3.2 Gram-Schmidt Orthogonalization	13/13
3.3 Householder QR Factorization	18/18

Week 4: Linear Least Squares	Score
4.1 Opening Remarks	2/2
4.2 Solution via the Method of Normal Equations	13/13
4.3 Solution via the SVD	7/7
4.4 Solution via the QR Factorization	5/5

Week 5: The LU and Cholesky Factorizations	Score
5.1 Opening Remarks	3/3
5.2 From Gaussian elimination to LU factorization	16/16
5.3 LU factorization with (row) pivoting	17/17
5.4 Cholesky factorization	11/11

Week 6: Numerical Stability	Score
6.1 Opening Remarks	2/2
6.2 Floating Point Arithmetic	14/14
6.3 Error Analysis for Basic Linear Algebra Algorithms	10/10
6.4 Error Analysis for Solving Linear Systems	12/12

Week 7: Sparse linear systems	Score
7.1 Opening Remarks	2/2
7.2 Direct solution	8.9375/9
7.3 Iterative Solution	11/11

Week 8: Descent Methods	Score
8.1 Opening	1/1

Week 8: Descent Methods	Score
8.2 Search directions	11/11
8.3 The Conjugate Gradient Method	10/10

Week 9: Eigenvalues and Eigenvectors	Score
9.1 Opening	2/2
9.2 Basics	26/26
9.3 The Power Method and related approaches	10/10

Week 10: Practical Solution of the Hermitian Eigenvalue Problem	Score
10.1 Opening	5/5
10.2 From Power Method to a simple QR algorithm	15/15
10.3 A Practical Hermitian QR Algorithm	11/11

Week 11: Computing the SVD	Score
11.1 Opening	3/3
11.2 Practical Computation of the Singular Value Decomposition	9/9
11.3 Jacobi's Method	6/6

For progress on ungraded aspects of the course, view your Course Outline.

Media Kit